

WASP-12b

R-1861

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_160121 · created 2026-08-03T09:29:48+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457407.78832 +/- 0.00024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.044
Transit depth (Rp/Rs)^2	1.2e-05 +/- 0.0031 [%]
Orbital Inclination (inc)	85.97 +/- 2.88
Airmass coefficient 1 (a1)	2.12 +/- 1.26
Airmass coefficient 2 (a2)	-0.6 +/- 0.87
Scatter in the residuals of the lightcurve fit is	68.53 %
Best Comparison Star	#1 - [507, 161]
Optimal Aperture	4.89
Optimal Annulus	5.7
Transit Duration (day)	0.1168 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.044 vs 0.1178 (deviation 2.49σ)
Duration fit vs published	0.1168 d vs 0.125 d (deviation 1.44σ)
Mid-time O-C	-1571.7 min
Depth SNR	0.0
Residual scatter	68.53 %

Light curve

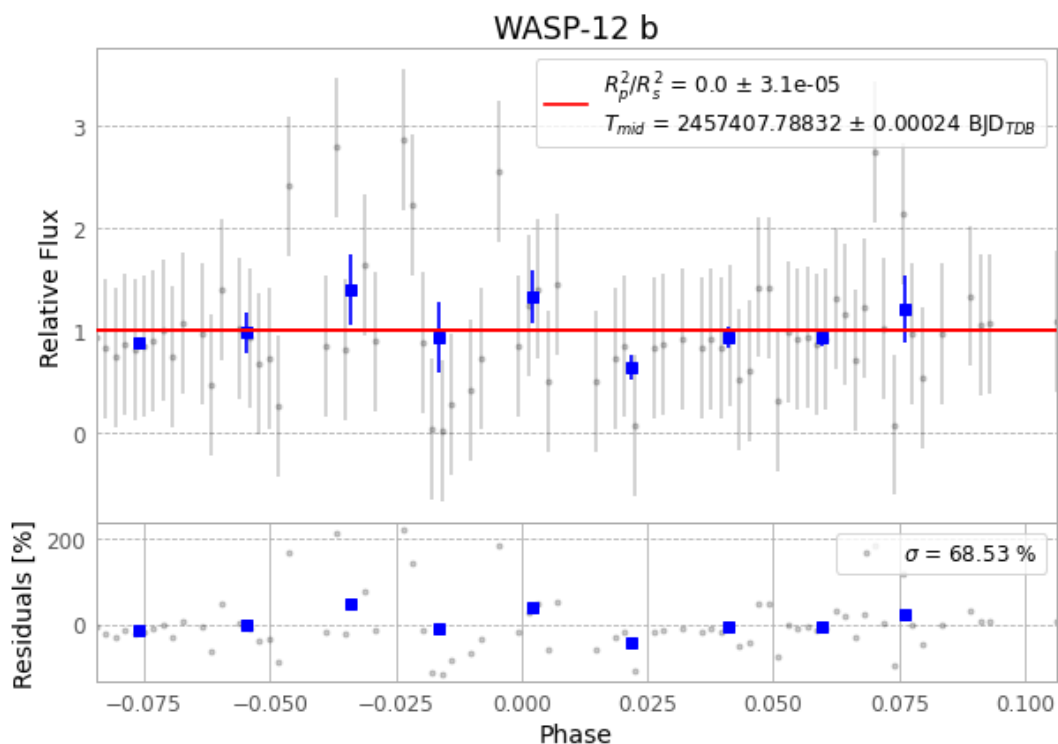


Figure 1 — FinalLightCurve_WASP-12 b_2016-01-19.png (SHA-256 48b5326eb0e19a64...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [416, 253], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 2a4e2111efea4e7d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

101 FITS frames (66 MB), WASP-12160120043313.FITS → WASP-12160120093312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-160121.log (7ba000b6c9d5...), FinalLightCurve_WASP-12 b_2016-01-19.png (48b5326eb0e1...), FinalLightCurve_WASP-12 b_2016-01-19.csv (2c7e3de112d6...)

Compute resources

Wall time 450.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 09:29Z.